Facial Expression Recognition with Image and Landmarks

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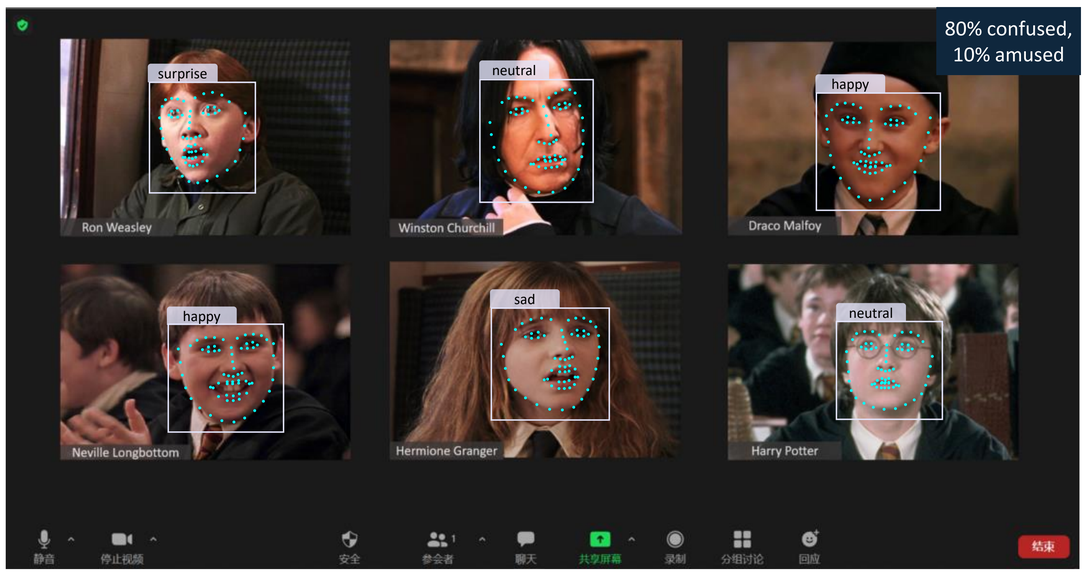

Figure 1. An ideal Zoom system informs lecturers of the students' real-time emotional responses. Input: Image of the gallery view captured by Zoom. Output: Emotion prediction of each person in the room.

Motivation

Lecturing via Zoom, though was only a backup plan during the Covid19 pandemic, has become widely used as an effective way of teaching now. Nevertheless, one problem with the current Zoom is that it is hard for the lecturer to timely get the emotional feedback from the students.

As we know, when a lecturer is immersed in illustrating an intriguing concept or tricky method, the students' display of interest or confusion will help the instructor make timely adjustments of the lecture pace and the method of illustrating knowledge. However, when an instructor is sharing the screen, he could not see the students' responses. On the other hand, if the instructor open the gallery view to see the students' faces, part of the screen will become invisible to the students.

Therefore, we design an ideal Zoom system as shown in Fig. 1 where its functions are optimized in teaching scenarios. Not only can the instructors see their shared slides, but also can they see the immediate prompt of the student's responses, such as the percent of the students becoming bored, detached, or excited at the top right corner of the screen.

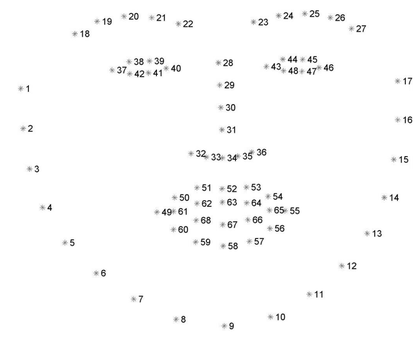

Figure 2. The landmark labels.

Workflow and method

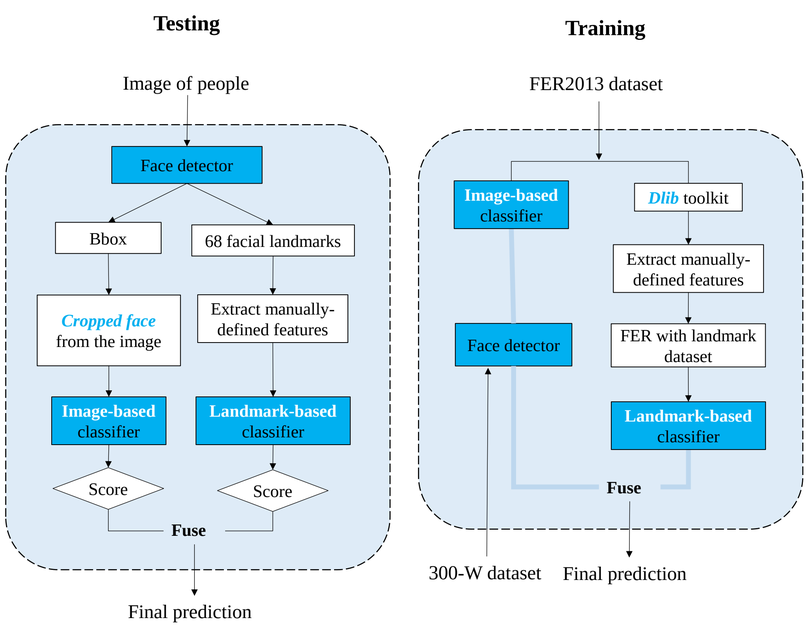

Figure 3. Training and testing workflow.

Procedure of testing:

1. Capture the zoom gallery view
2. Input the screenshot to the face detector and get each face's bounding boxes and 68 landmarks
3. Cut the faces as inputs to the *image-based classifier*
4. Extract features using landmarks as inputs to *landmark-based classifier*
5. Fuse the scores of the two classifiers and get the final prediction

Face detector

Dataset	size
LFPW	1035 images
HELEN	2330 images
AFW	337 images
IBUG	135 images
LS3D-W	230,000 images

Table 1. Dataset for face detector [1] [2].

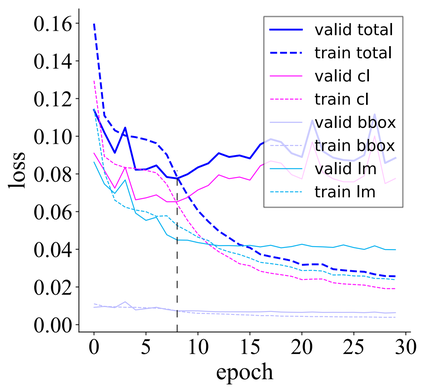

Figure 4. Loss during training.

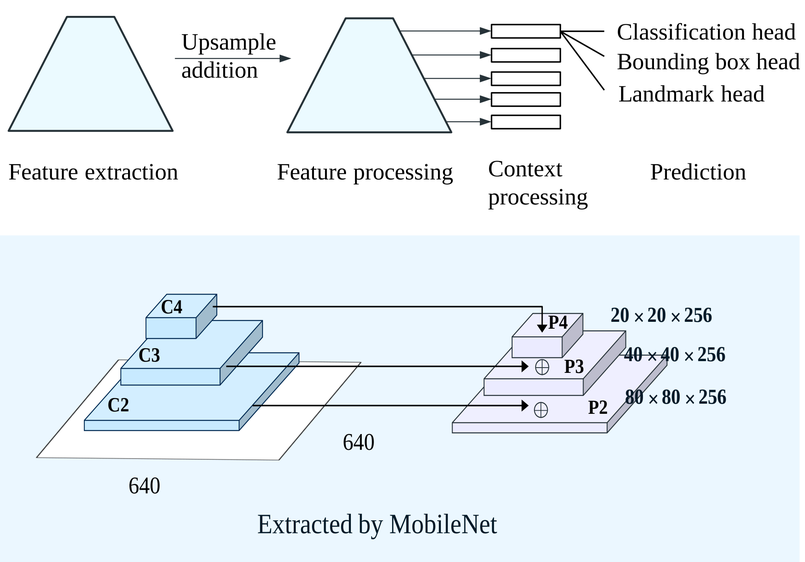

Figure 5. Structure of RetinaFace. **Upper:** The three components of the structure, including the feature pyramid module, context processing module, and the prediction module. **Bottom:** Feature pyramid extracted by MobileNet.

	Heavyweight version	Lightweight-our choice
Backbone	ResNet-152 [4]	MobileNet [5]
Model size	262 MB	1.54 MB
Feature pyramid	5 layers	3 layers
#landmark	5	68

Table 2. Comparison of the heavyweight version and the lightweight version (our choice) of *RetinaFace* [3]. The differences include a backbone of MobileNet, a 3-layer feature pyramid and the modified landmark heads which produces 68 landmarks.

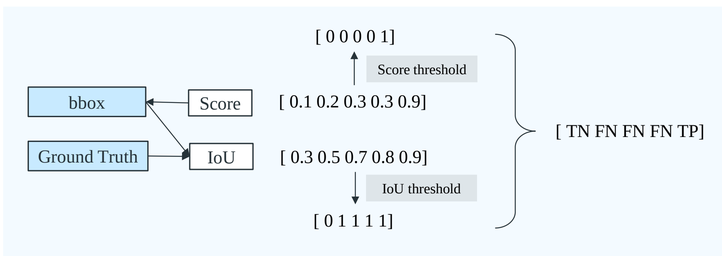

Figure 6. Method of classifying a prediction into TN, FN, TP, FP.

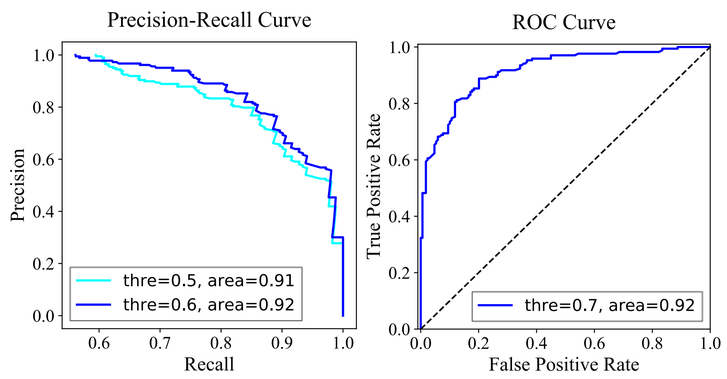

Figure 7. Precision-Recall curve and ROC curve. **Left:** PR curve under the thresholds of 0.5 and 0.6. **Right:** ROC curve under the threshold of 0.7.

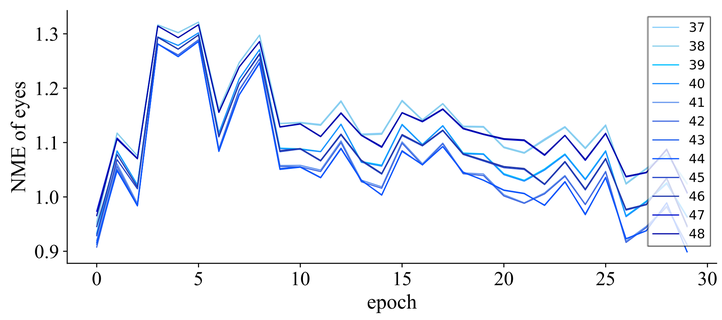

Figure 8. Normalized mean error (NME) of eye landmarks during training. Since the NME of all the displayed key points displays a stable tendency of decreasing, our model is learning effectively.

Image-based Classifier

FER 2013						
angry	disgust	fear	happy	neutral	sad	surprise
4953	547	5121	8989	6198	6077	4002

Table 3. We train the Emotion classifier 1 on the FER2013 [6] dataset, which contains **35887** grayscale images of size **48×48** with **7** emotion classes.

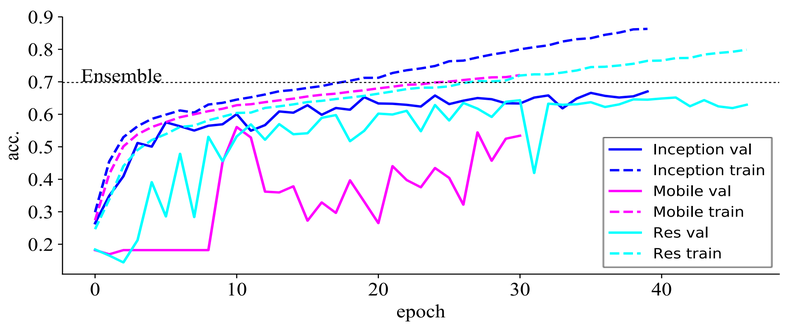

Figure 9. Accuracy of three image-based classifiers. MobileNet is very susceptible to overfitting, and InceptionNet and ResNet performs better.

Landmark-based Classifier

Face expression is closely related to the movement of muscles. We defined a distance feature by landmark information to describe the movement of muscles in some facial part (eyebrows, eyes, and mouth):

1. Eyebrows (normalized by the distance of $P_{22} - P_{40}$ or $P_{23} - P_{43}$)
2. Eyes (normalized by the width of an eye)
3. Mouth (normalized by the height of mouth)

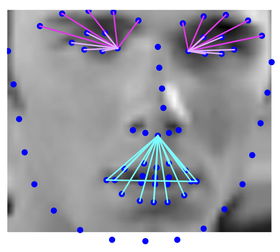

Figure 10. Defined features.

The input feature vector has 176 dimensions with 68 landmark coordinates and self-defined distance vector. We adopt a MLP model with 2 hidden layers. One has 176 neurons and the other has 512 neurons.

Model fusion

Model	Size	Num of para.	Test Acc.
MobileNet V2	14 M	4,253,864	55.1%
ResNet50	98 M	25,636,712	63.5%
Inception V3 [7]	92 M	23,851,784	68.0%
MLP	606 K	125,367	60.0%
Ensemble	/	/	71.7% for 7 classes, 82.9% for 3 classes

Table 4. Specification of the image-based classifier and the landmark-based classifier.

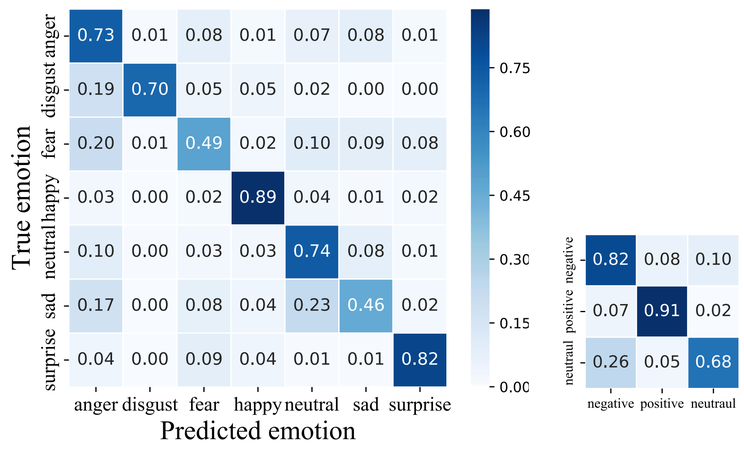

Figure 11. Confusion matrix of the fused model. **Left:** 7 classes. **Right:** 3 classes.

Conclusion

In conclusion, we innovatively design a real-time emotion reminder embedded in Zoom system, which will help the professor have a clearer picture of how the class goes and make intermediate adjustments. To achieve this, we define a multi-people emotion analysis task and propose a three-stage workflow. Moreover, for the task of emotion recognition, we use detected landmarks to extract important facial features such as the angle of smile and eyes, which enhance the performance.

Reference

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[7] Chollet, François. "Xception: Deep learning with depthwise separable convolutions." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.